

**MAR ATHANASIOUS COLLEGE OF ENGINEERING
KOTHAMANGALAM, KERALA – 686 666**

(Government Aided Autonomous Institution)



Department of Computer Applications

MINI PROJECT REPORT

on

**MEDICAL SPECIALTY CLASSIFICATION USING
TF-IDF AND ENSEMBLE MACHINE LEARNING**

Submitted by

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Under the Guidance of

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Department of Computer Applications

In partial fulfilment for the award of the degree of
Master of Computer Applications

APJ Abdul Kalam Technological University, Thiruvananthapuram

OCTOBER 2026

MAR ATHANASIOUS COLLEGE OF ENGINEERING
KOTHAMANGALAM, KERALA – 686 666
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CERTIFICATE



**MEDICAL SPECIALTY CLASSIFICATION USING TF-IDF AND
ENSEMBLE MACHINE LEARNING**

Certified that this is the bonafide record of the Mini Project work carried out by

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during the third semester, in partial fulfilment of the requirements for the award of the degree of

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DECLARATION

I, Jiphin George, student of Master of Computer Applications, Department of Computer Applications, Mar Athanasius College of Engineering, Kothamangalam, hereby declare that this Mini Project Report titled “**Medical Specialty Classification using TF-IDF and Ensemble Machine Learning**” is a bonafide record of the academic project work carried out by me under the supervision and guidance of **Prof. Biju Skaria**, Project Guide, Department of Computer Applications, in partial fulfilment of the requirements for the award of the degree of Master of Computer Applications of APJ Abdul Kalam Technological University, Thiruvananthapuram.

I further declare that this report has not been submitted previously, either in part or in full, to any other University or Institution for the award of any degree or diploma.

Place: Kothamangalam

Date: 30 October 2026

Jiphin George

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ABSTRACT

The rapid digitisation of modern healthcare has led to an exponential influx of unstructured clinical text within Electronic Health Record (EHR) systems, including operative summaries, consultation notes, and hospital discharge reports. Manually indexing and triaging these clinical narratives into specialized medical departments is labor-intensive, error-prone, and bottlenecked by severe inter-annotator variability.

This project designs, evaluates, and deploys an automated clinical decision-support platform, **MedSpecialty AI**, for clinical text categorization into medical specialties using high-dimensional Term Frequency–Inverse Document Frequency (TF-IDF) feature engineering and an Ensemble Machine Learning architecture. The investigation utilizes the MTSamples clinical transcription corpus comprising 4,999 raw records across 40 categories. Systematic exploratory data analysis revealed significant label overlap and procedural ambiguity in broad administrative classes (e.g., General Surgery, Consult notes). To construct an academically rigorous and clinically realistic benchmark, a curated dataset of 1,663 records across eight mutually exclusive hospital departments was established: Cardiovascular / Pulmonary, Orthopedic, Gastroenterology, Neurology, Urology, Obstetrics / Gynecology, ENT – Otolaryngology, and Ophthalmology.

The natural language processing pipeline executes text normalization, medical punctuation handling, tokenization, stopword removal, morphological lemmatization, and sublinear TF-IDF vectorization across an 8,000-dimensional unigram and bigram vocabulary (L_2 normalized). Four candidate classifiers with diverse inductive biases were trained on stratified partitions (80% train / 20% test, 333 holdout records): Linear Support Vector Machine (Linear SVM with Platt probability calibration), Logistic Regression, Random Forest, and Multinomial Naive Bayes. Empirical evaluation demonstrates that Linear SVM achieved the highest individual classification accuracy of 88.59% with a Macro F1-score of 0.8971. A Soft Voting Ensemble combining all four models with calibrated weights [2:2:1:1] achieved 87.99% accuracy and a Macro F1-score of 0.8898, delivering well-calibrated posterior probability spreads.

To ensure patient safety in open-world clinical environments, the system incorporates an **Open-Set Confidence Rejection Gate** ($\tau = 0.48$). Any transcription exhibiting peak ensemble confidence below 48% is routed to a 9th operational class, “*Other / Requires Review*”, successfully intercepting unsupported clinical specialties (such as Dermatology, Psychiatry, or Dentistry)

and non-medical text without false specialty force-fitting. The complete pipeline is operationalized through a responsive web platform featuring in-memory PDF clinical text extraction via PyMuPDF, interactive 7-phase dataflow visualization, TF-IDF feature space inspection, and dynamic multi-model consensus dashboards.

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1. INTRODUCTION

1.1 Background

In contemporary healthcare institutions, the widespread adoption of Electronic Health Record (EHR) and Electronic Medical Record (EMR) systems has resulted in the continuous generation of massive volumes of clinical narratives. Healthcare practitioners generate extensive unstructured text daily, including admission notes, preoperative consultations, surgical operative summaries, progress notes, and diagnostic evaluations. Accurately categorizing these unstructured clinical notes into specific medical specialties (such as Cardiology, Neurology, Orthopedics, Gastroenterology, Urology, and Ophthalmology) is an indispensable prerequisite for hospital workflow automation, patient file routing, diagnostic database indexing, and clinical decision support.

Traditionally, document indexing relies on manual review by medical coders and health information management personnel. With escalating patient volumes in modern medical centers, manual categorization has become a severe administrative bottleneck, introducing substantial latency, high labor costs, and inter-annotator classification discrepancies. Natural Language Processing (NLP) paired with Machine Learning (ML) provides an automated computational paradigm capable of rapidly extracting semantic signals from clinical narratives, standardizing specialty categorization, and routing medical documentation in real time.

1.2 Problem Statement

Manual categorization of medical transcription narratives into appropriate clinical specialties is labor-intensive, time-consuming, and prone to subjective diagnostic interpretation. Unstructured clinical notes present formidable linguistic complexities, including dense domain-specific medical terminology, non-standard abbreviations, variable document lengths, and anatomical lexical overlap across adjacent clinical sub-disciplines.

Furthermore, authentic medical transcription datasets suffer from severe class imbalance and noisy labeling. Datasets such as the Kaggle Medical Transcriptions (MTSamples) corpus initially group notes into broad procedural or administrative categories (such as General Surgery, SOAP Notes, and Consultative Summaries), which overlap heavily with organ-specific departments. A critical limitation of conventional machine learning systems is their closed-set

assumption: they assume that every incoming clinical record belongs to one of the pre-defined target classes. In real-world hospital environments, out-of-scope clinical narratives (such as Dermatology, Psychiatry, or Pediatrics) or non-medical text are routinely encountered. Closed-set classifiers mistakenly force-fit these out-of-domain cases into an arbitrary specialty, compromising clinical integrity. Additionally, deep learning transformer models (such as BERT or BioBERT) incur excessive GPU computational overhead, rendering them impractical for lightweight, instantaneous web deployment.

1.3 Need for the System

Automating medical specialty classification through efficient statistical natural language processing and ensemble machine learning addresses critical clinical needs:

- *Streamlined Clinical Triage*: Eliminates manual document sorting backlogs, enabling automated indexing and immediate department routing upon clinical dictation.
- *Rapid Emergency Retrieval*: Empowers attending physicians to quickly access historical specialty-specific operative notes during acute patient emergencies.
- *Open-Set Patient Safety*: Implements a confidence-gated rejection mechanism that identifies out-of-scope or ambiguous medical documents, safely routing them to manual clinical review rather than producing hazardous misclassifications.
- *Explainable NLP Representation*: Provides algorithmic transparency through Term Frequency–Inverse Document Frequency (TF-IDF) feature attribution, allowing clinicians to inspect the exact keywords driving model decisions.
- *Computationally Efficient Web Deployment*: Delivers instantaneous inference on standard CPU architectures through lightweight classical estimators, hosted via an accessible web platform with direct PDF report ingestion.

1.4 Objectives

The primary objective of this project is to design, develop, and deploy a robust, end-to-end automated Medical Specialty Classification system (**MedSpecialty AI Platform**) using sublinear TF-IDF vectorization and an Ensemble Machine Learning architecture with open-set rejection capabilities. The specific objectives are:

1. To conduct comprehensive Exploratory Data Analysis (EDA) on the raw MTSamples medical transcriptions corpus (4,999 records across 40 categories) and identify noisy, overlapping, and administrative document classes.
2. To curate a clinically sound, mutually exclusive 8-specialty benchmark dataset comprising 1,663 records across core hospital departments.
3. To build a modular clinical NLP preprocessing pipeline incorporating lowercasing, medical punctuation handling, tokenization, stopword removal, and morphological lemmatization.
4. To extract discriminative unigram and bigram features using an 8,000-dimensional sub-linear TF-IDF vectorizer with L_2 Euclidean normalization.
5. To train and evaluate candidate supervised classifiers with distinct inductive biases: Linear Support Vector Machine (Linear SVM with Platt probability calibration), Logistic Regression, Random Forest, and Multinomial Naive Bayes.
6. To construct a Soft Voting Ensemble Classifier utilizing calibrated weights [2:2:1:1] to synthesize probabilistic consensus across candidate models.
7. To formulate an Open-Set Confidence Rejection mechanism with a calibrated safety threshold ($\tau = 0.48$) establishing a 9th operational class ("*Other / Requires Review*") for out-of-scope clinical records.
8. To develop and deploy an interactive, production-grade web application using Flask, supporting drag-and-drop PDF text extraction via PyMuPDF, real-time transcription editing, and 7-phase dataflow visualization.

1.5 Scope

The scope of this project encompasses:

- *Dataset Scope*: Evaluates 4,999 raw clinical transcription records from the Kaggle MT-Samples corpus, filtering out non-specialty administrative document formats (SOAP notes, Consults, Discharge Summaries) and overlapping umbrellas (General Surgery) to establish a curated benchmark of 1,663 records across eight core hospital departments:

1. Cardiovascular / Pulmonary
 2. Orthopedic
 3. Gastroenterology
 4. Neurology
 5. Urology
 6. Obstetrics / Gynecology
 7. ENT – Otolaryngology
 8. Ophthalmology
- *Feature Engineering Scope:* Focuses on unigram and bigram TF-IDF representations bounded at 8,000 features, applying logarithmic term frequency dampening ($1 + \log(\text{TF})$) to prevent high-frequency term domination.
 - *Algorithmic Scope:* Investigates four classical machine learning models (Linear SVM, Logistic Regression, Random Forest, and Multinomial Naive Bayes), a Soft Voting Ensemble, and a threshold-based open-set rejection gate.
 - *Deployment Scope:* Delivers a fully responsive web application (**MedSpecialty AI Platform**) built with Flask and Vanilla CSS/JavaScript, featuring in-memory PDF parsing (PyMuPDF) and full pipeline explainability without requiring external GPU infrastructure.

1.6 Project Overview

The project follows a structured engineering workflow:

1. *Data Ingestion & Exploratory Profiling:* Loading raw MTSamples CSV data, analyzing missing attributes, document length distributions, and category imbalances.
2. *Dataset Curation & Cleaning:* Pruning null records, resolving category ambiguities, and extracting enriched clinical text (combining consultation abstracts and transcription narratives).
3. *NLP Preprocessing:* Standardizing text via regex cleaning, lowercasing, punctuation stripping, tokenization, stopwords removal, and WordNet morphological lemmatization.

4. *Feature Extraction*: Vectorizing preprocessed narratives into an 8,000-dimensional sparse feature space using sublinear TF-IDF scaling.
5. *Model Development & Calibration*: Training candidate estimators on stratified training partitions (80% train / 20% test) and applying Platt scaling via `CalibratedClassifierCV` to Linear SVM.
6. *Ensemble Integration*: Aggregating model posteriors using weighted soft voting $(2 \times \text{SVM} + 2 \times \text{LR} + 1 \times \text{RF} + 1 \times \text{MNB})/6$.
7. *Open-Set Confidence Gating*: Applying decision threshold $\tau = 0.48$ to classify cases into confirmed specialties or route them to “*Other / Requires Review*”.
8. *Web Platform Deployment*: Serializing artifacts with Joblib and serving predictions via the MedSpecialty AI Platform with PDF ingestion.

1.7 Expected Outcome

The project delivers the following primary outcomes:

- A curated, high-quality benchmark dataset of 1,663 medical transcriptions across eight clinical specialties.
- An empirical comparative analysis confirming that Linear SVM achieves the highest individual accuracy (88.59%, Macro F1: 0.8971) and that the Soft Voting Ensemble achieves superior probability calibration (87.99%, Macro F1: 0.8898).
- A verified open-set rejection mechanism that accurately routes unsupported clinical records (e.g., Dermatology, Psychiatry) to the “*Other*” category.
- A fully functional, production-ready web application providing real-time PDF extraction and clinical NLP explainability.
- Comprehensive academic documentation adhering to MCA departmental standards.

2. SUPPORTING LITERATURE

2.1 Literature Review

A systematic review of contemporary literature was conducted to evaluate existing methodological frameworks, feature engineering representations, classification algorithms, class imbalance mitigation techniques, and dataset challenges in clinical natural language processing. The review focuses on peer-reviewed research analyzing medical narratives, electronic health records, and document categorization systems.

Paper 1 – Omar AM, Elhoseny M, Hassanien AM. Clinical text classification with word representation features and machine learning algorithms. International Journal of Online and Biomedical Engineering (iJOE). 2023.

The paper titled “Clinical Text Classification with Word Representation Features and Machine Learning Algorithms” [1] investigates the impact of diverse text representation techniques on Electronic Medical Record (EMR) narratives. The authors introduced a structured four-phase classification framework consisting of clinical text preprocessing, word vectorization, dimensionality reduction, and supervised classification.

To model clinical narratives, the study conducted a comparative assessment of classical Bag-of-Words (BoW) and Term Frequency–Inverse Document Frequency (TF-IDF) representations against dense distributed semantic embeddings (Word2Vec) across multiple supervised estimators, including Logistic Regression, Support Vector Machine (SVM), Naive Bayes, and k-Nearest Neighbors (k-NN). The empirical evaluations indicated that pairing continuous Word2Vec embeddings with a k-NN classifier yielded the highest classification accuracy of 92%, demonstrating that continuous representations capture semantic affinities between medical terms more effectively than unigram Bag-of-Words.

However, the authors noted that distance-based k-NN introduces significant computational latency during test-time inference, scaling poorly to large-scale hospital databases. Furthermore, the investigation evaluated individual classifiers in isolation without exploring multi-model ensemble consensus or real-time web deployment frameworks. Crucially, Principal Component Analysis (PCA) was evaluated exclusively as an offline dimensionality reduction step in Paper 1 and is not required in our proposed architecture, as sublinear TF-IDF with feature pruning

maintains optimal sparsity without dense projection.

TITLE	Clinical Text Classification with Word Representation Features and Machine Learning Algorithms [1]
AREA OF WORK	Clinical Natural Language Processing / EMR Text Vectorization
DATASET	Electronic Medical Record (EMR) Clinical Transcriptions Dataset
METHODOLOGY / STRATEGY	4-Phase Framework: (1) Preprocessing, (2) Vectorization (BoW, TF-IDF, Word2Vec), (3) Feature Reduction (PCA), (4) Supervised Classification
ALGORITHM	Logistic Regression, Support Vector Machine (SVM), Naive Bayes, k-NN, PCA
RESULT / PRECISION	Word2Vec + k-NN achieved the highest classification accuracy of 92%
ADVANTAGES	Proves dense semantic representations capture clinical context better than raw unigram BoW
LIMITATIONS	Distance-based k-NN exhibits high inference latency; lacks ensemble voting, open-set handling, and real-time web deployment
RELEVANCE TO MY PROJECT	Validates the standard 4-phase clinical text preprocessing sequence and benchmark machine learning estimators
FUTURE PROPOSAL	Develop lightweight ensemble classifiers suitable for instantaneous CPU web inference with calibrated confidence

Table 1: Summary of Paper 1

Paper 2 – Zhang H, Zhu D, Tan H, Shafiq M, Gu Z. Medical specialty classification based on semiadversarial data augmentation. Hindawi Computational Intelligence and Neuroscience. 2023.

The research article titled “Medical Specialty Classification Based on Semiadversarial Data Augmentation” [2] investigated extreme category imbalance and data sparsity in clinical specialty categorization. The authors introduced Semiadversarial Data Augmentation (SemiADA), a framework generating synthetic clinical text samples along adversarial perturbation trajectories to populate sparse decision regions in high-dimensional feature spaces.

Additionally, a Probabilistic Information (PI) layer was incorporated following the Softmax output to recalculate prediction confidence by weighting domain-specific clinical nouns. The authors evaluated a filtered subset of the Kaggle Medical Specialty Classification dataset re-

stricted to 18 classes, achieving an average performance improvement of 14.9% in accuracy and F1-score over baseline benchmarks. Their findings confirmed that domain-specific clinical nouns carry the primary discriminative signal for medical department routing.

Despite these gains, generating synthetic samples along adversarial attack paths incurs extreme computational overhead. Furthermore, fine-tuning deep transformer architectures (such as BERT or BioBERT) requires dedicated GPU clusters and introduces substantial inference latency, creating a major barrier for real-time clinical web applications operating in resource-constrained hospital environments. The study also operated strictly under a closed-set assumption, omitting mechanisms to intercept out-of-domain records.

TITLE	Medical Specialty Classification Based on Semiadversarial Data Augmentation [2]
AREA OF WORK	Clinical Text Data Augmentation / Severe Class Imbalance Mitigation
DATASET	Kaggle Medical Specialty Classification Dataset (Subset restricted to 18 classes)
METHODOLOGY / STRATEGY	Semiadversarial sample generation along attack trajectories + Probabilistic Information (PI) noun-weighted Softmax recalculation
ALGORITHM	SemiADA, BERT (Transformer), Softmax with PI Noun Layer
RESULT / PRECISION	Achieved an average 14.9% boost in Accuracy and Macro F1-score over baseline benchmarks
ADVANTAGES	Populates sparse minority decision boundaries using geometry-aware synthetic clinical samples
LIMITATIONS	Extreme training latency; fine-tuning BERT demands dedicated GPU accelerators; closed-set classification only
RELEVANCE TO MY PROJECT	Highlights class imbalance and noun significance in MTSamples data; motivates high-dimensional TF-IDF with ensemble ML
FUTURE PROPOSAL	Combine sublinear TF-IDF n-gram weighting with probability-calibrated ensemble learning and open-set rejection on standard CPUs

Table 2: Summary of Paper 2

Paper 3 – Blanchard AE, Gao S, Yoon HJ, Tourassi G. A keyword-enhanced approach to handle class imbalance in clinical text classification. IEEE Journal of Biomedical and Health Informatics. 2022.

Blanchard et al., writing in the *IEEE Journal of Biomedical and Health Informatics* [3], addressed the challenge of deep neural networks overfitting on rare clinical categories when trained on lengthy, complex pathology narratives. The authors introduced a data-centric regularisation strategy by injecting short class-associated keyword sequences into mini-batches during neural network training, formulating a dual loss function:

$$L = L_{\text{docs}} + \alpha L_{\text{key}} \quad (1)$$

where L_{docs} evaluates cross-entropy loss over full documents, L_{key} penalizes misclassifications on isolated diagnostic keyword sequences, and α balances the objective.

Evaluating on the National Cancer Institute (NCI) Surveillance, Epidemiology, and End Results (SEER) cancer pathology registry dataset, the study benchmarked Convolutional Neural Networks (CNNs), keyword-constrained loss functions, and conventional resampling techniques (oversampling and undersampling). The findings demonstrated that keyword constraints dramatically improved Macro F1-score and minority class recall without degrading majority class performance. Crucially, the authors emphasized that overall micro-accuracy is an inherently deceptive evaluation metric in clinical NLP, as high global accuracy frequently masks near-zero recall on rare clinical conditions.

However, their approach relies on manual curation of comprehensive keyword lexicons by clinical domain experts for every target category, limiting rapid scalability to new medical specialties. This limitation directly motivates our system’s automated statistical TF-IDF approach, which extracts informative unigram and bigram vocabulary automatically without requiring hand-crafted dictionaries.

TITLE	A Keyword-Enhanced Approach to Handle Class Imbalance in Clinical Text Classification [3]
AREA OF WORK	Imbalanced Clinical Narrative Categorization / Pathology NLP
DATASET	NCI SEER Surveillance Cancer Pathology Registry Reports Dataset
METHODOLOGY / STRATEGY	Mini-batch keyword sequence injection to constrain the neural loss function and prevent rare-class forgetting
ALGORITHM	Convolutional Neural Networks (CNNs), Keyword-Constrained Loss Optimization
RESULT / PRECISION	Significantly boosted Macro F1-score and minority recall across imbalanced pathology classes
ADVANTAGES	Demonstrates that keyword and n-gram terms carry decisive clinical signals; confirms Macro F1 as the primary clinical metric
LIMITATIONS	Requires exhaustive manual curation of keyword dictionaries by medical specialists; computationally heavy neural training
RELEVANCE TO MY PROJECT	Proves that n-gram keyword weighting preserves rare class discrimination; validates Macro F1 as our primary evaluation metric
FUTURE PROPOSAL	Implement automated statistical TF-IDF unigram and bigram extraction with sublinear scaling, eliminating manual dictionary construction

Table 3: Summary of Paper 3

2.2 Literature Review Summary

Table 4 presents a comparative synthesis of the reviewed literature alongside our proposed MedSpecialty AI Platform.

Paper	Area	Dataset	Algorithm	Advantages	Relation to Proposed Work
Omar et al. [1]	Clinical Text Vectorization	EMR Text	LR, SVM, NB, k-NN	High accuracy (92%) with Word2Vec	Establishes the 4-phase text preprocessing pipeline
Zhang et al. [2]	Data Augmentation	MTSamples (18 classes)	SemiADA, BERT	+14.9% gain in Accuracy and F1	Identifies MTSamples class imbalance and clinical noun signal
Blanchard et al. [3]	Class Imbalance Regularization	NCI SEER Pathology	CNN + Keyword Loss	High Macro F1 on minority classes	Establishes Macro F1 as primary metric; validates n-gram term weighting
Proposed Work	Specialty Triage & Open-Set Routing	MTSamples (Curated 8 Core + Other)	Sublinear TF-IDF (8k) + Soft Voting Ensemble [2:2:1:1]	Fast CPU inference, open-set safety gate ($\tau = 0.48$), interactive web UI	Complete, verified clinical decision support platform (MedSpecialty AI)

Table 4: Overall Summary of Literature and Relation to Proposed System

Evolution of Research

Clinical natural language processing has evolved from elementary Bag-of-Words term frequency counters toward distributed word embeddings (Word2Vec), deep convolutional architectures, complex transformer models (BERT, BioBERT), and adversarial data augmentation (SemiADA). While deep neural networks deliver high semantic abstraction, their steep computational overhead, large parameter footprints, and GPU hardware prerequisites prevent real-time deployment in cost-conscious clinical web environments. Furthermore, deep neural networks often suffer from overconfidence on out-of-domain text.

Identified Research Gaps

The literature review reveals four fundamental research gaps:

1. *Excessive Computational Overhead*: Existing deep learning systems rely on GPU hardware accelerators and introduce substantial inference latency, making real-time interactive dictation triage challenging.
2. *Closed-Set Vulnerability*: Prior works operate exclusively under a closed-set assumption, failing to handle out-of-scope medical specialties or non-clinical narratives, resulting in hazardous false specialty assignments.
3. *Lack of End-to-End Clinical Deployment*: Most studies focus solely on offline experimental benchmarks without delivering an operational, accessible web application supporting direct PDF ingestion.
4. *Single Estimator Bias*: Many prior systems rely on a single classifier architecture, failing to exploit ensemble diversity to balance high-margin boundaries with calibrated probabilistic spreads.

Motivation for Proposed System

Our project directly addresses these research gaps by designing the **MedSpecialty AI Platform**. The system couples an 8,000-dimensional sublinear TF-IDF vectorizer (capturing unigram terms and bigram clinical phrases) with a calibrated Soft Voting Ensemble aggregating Linear SVM, Logistic Regression, Random Forest, and Multinomial Naive Bayes using optimal weights [2:2:1:1]. Crucially, an Open-Set Confidence Rejection Gate ($\tau = 0.48$) is established to route unsupported or ambiguous narratives to a dedicated “*Other / Requires Review*” class. The entire architecture is deployed through a responsive Flask web application featuring PyMuPDF in-memory text parsing and real-time NLP pipeline inspection on standard CPU hardware.

2.3 Findings and Proposals

2.3.1 Findings

The systematic review of contemporary literature yields several critical findings:

1. *Efficacy of Classical ML with N-Gram Term Weighting*: Classical machine learning algorithms (Support Vector Machines, Logistic Regression, Random Forest, and Naive Bayes) demonstrate competitive classification performance on clinical narratives when paired with sublinear TF-IDF feature representations, matching or exceeding heavy neural models on moderate-sized corpora while executing in milliseconds on standard CPUs.
2. *Sublinear TF Dampening Prevents Frequency Bias*: Clinical dictations feature frequent repetitions of routine procedural terms. Sublinear logarithmic scaling ($1 + \log(\text{TF})$) successfully prevents high-frequency terms from overshadowing specialty-defining diagnostic keywords.
3. *Complementary Estimator Inductive Biases*: Different classifier families exhibit complementary strengths. Linear SVM identifies robust maximum-margin hyperplanes in high-dimensional sparse spaces; Logistic Regression provides well-calibrated odds; Random Forest captures non-linear feature interactions; and Multinomial Naive Bayes contributes probabilistic frequency baselines.
4. *Necessity of Open-Set Rejection*: Closed-set classification in clinical environments creates unacceptable diagnostic risks when processing out-of-scope conditions. An empirical confidence threshold provides an effective safeguard for triaging unrecognized narratives to human experts.
5. *Importance of Macro F1 Evaluation*: In clinical NLP with imbalanced specialty distributions, overall accuracy is deceptive. Macro F1-score—which weights each clinical department equally—serves as the primary indicator of balanced diagnostic capability.

2.3.2 Proposal

Synthesizing these findings, this project proposes the **MedSpecialty AI Platform** for automated clinical specialty classification and open-set triage:

1. *Curated Benchmark Dataset*: Filter the raw MTSamples corpus down from 40 overlapping/administrative categories to a curated benchmark of 1,663 records across eight mutually exclusive hospital specialties: Cardiovascular / Pulmonary, Orthopedic, Gastroenterology, Neurology, Urology, Obstetrics / Gynecology, ENT – Otolaryngology, and Ophthalmology.

2. *Modular NLP Preprocessing*: Implement an automated cleaning pipeline performing lowercasing, regex noise removal, tokenization, stopwords elimination, and WordNet morphological lemmatization.
3. *Sublinear TF-IDF Vectorization*: Extract unigram and bigram features bounded at 8,000 dimensions with sublinear term frequency scaling, document frequency cutoffs ($\text{min_df} = 2, \text{max_df} = 0.70$), and Euclidean L_2 normalization.
4. *Calibrated Soft Voting Ensemble*: Train Linear SVM (calibrated via Platt scaling), Logistic Regression, Random Forest, and Multinomial Naive Bayes, synthesizing their predictions using calibrated soft voting weights $(2 \times \text{SVM} + 2 \times \text{LR} + 1 \times \text{RF} + 1 \times \text{MNB})/6$.
5. *Open-Set Rejection Architecture*: Formulate a decision rule: if the peak ensemble probability $P_{\max} < 0.48$, the document is classified as “Other” with status “Out of Scope / Requires Review”, displaying nearest candidate specialties for manual triage.
6. *Interactive Web Application*: Deploy a production-grade web application featuring in-memory PDF extraction (PyMuPDF), editable clinical dictation areas, interactive TF-IDF feature tables, and multi-model consensus dashboards.

3. SYSTEM ANALYSIS

System analysis establishes the foundational functional requirements, dataset architecture, exploratory statistical properties, preprocessing transformations, algorithmic mechanics, and deployment feasibility for the automated Medical Specialty Classification system.

The empirical investigation is conducted on the Kaggle Medical Transcriptions (MTSamples) dataset [4], which provides authentic, transcribed clinical dictations across diverse healthcare disciplines. Analyzing this corpus facilitates the development of a clinically sound, computationally lightweight natural language processing pipeline capable of differentiating domain-specific medical vocabulary while remaining suitable for real-time web deployment on standard CPU hardware.

3.1 Analysis of the dataset

3.1.1 About the dataset

Dataset Overview

The Kaggle Medical Transcriptions dataset is a curated collection of authentic clinical narratives obtained from MTSamples.com, an open clinical transcription repository created to assist medical transcriptionists and healthcare informatics researchers. The dataset encapsulates a broad spectrum of real-world clinical encounters, ranging from routine consultative histories and physical examinations to highly complex intraoperative surgical summaries.

- **Total Raw Records:** 4,999 clinical transcription reports.
- **Total Attributes:** 6 attributes (1 integer identifier, 4 textual features, and 1 categorical target label).
- **Data Types:** Free-text clinical narratives (transcription), diagnostic abstracts (description), comma-delimited keywords (keywords), procedure titles (sample_name), and categorical specialty labels (medical_specialty).
- **Target Attribute:** medical_specialty, containing 40 nominal medical department categories in the raw distribution.

Table 5 summarizes the fundamental structural characteristics of the raw dataset.

Property	Description
Dataset Name	Kaggle Medical Transcriptions (MTSamples) Dataset
Source	Kaggle (https://www.kaggle.com/datasets/tboyle10/medicaltranscriptions)
Total Raw Records	4,999 clinical reports
Number of Attributes	6 fields (Unnamed: 0, description, medical_specialty, sample_name, transcription, keywords)
Primary Feature Variable	transcription (Unstructured clinical narrative dictation)
Auxiliary Feature Variable	description (Short diagnostic synopsis)
Target Class Variable	medical_specialty (40 raw categories; curated to 8 core specialties + 1 Open-Set class)
Advantages	Authentic clinical language; zero missing target labels (100% labeled); rich multi-specialty terminology
Challenges	Severe category imbalance; missing transcription entries (33 nulls); overlapping administrative document categories

Table 5: Characteristics of the Kaggle Medical Transcriptions Dataset

Attribute Specifications

The six dataset attributes serve distinct analytical functions:

1. **Record Identifier (Unnamed: 0):** An integer sequence representing the raw tabular row index (range: 0 to 4,998).
2. **Clinical Abstract (description):** A concise, one-to-two sentence clinical synopsis summarizing the patient's presenting condition, consultation findings, or surgical objective.
3. **Target Medical Specialty (medical_specialty):** The categorical label designating the medical specialty or departmental division under which the transcription was categorized.
4. **Dictation Title (sample_name):** A brief descriptive title denoting the specific clinical procedure, anatomical site, or encounter type (e.g., "Coronary Artery Bypass", "Colonoscopy").

5. **Primary Clinical Narrative (transcription):** The central predictor feature containing the full, free-text clinical dictation. It details the patient’s medical history, physical examination, diagnostic investigations, intraoperative findings, and postoperative care plans.
6. **Extracted Keywords (keywords):** Comma-delimited medical keywords and anatomical descriptors indexed from the report.

3.1.2 Explore the dataset

Statistical Data Profiling and Null Value Audit

Exploratory data profiling was conducted using Python data analysis frameworks (pandas, numpy). Systematically inspecting the schema via shape, info(), dtypes, and null audits confirmed the presence of 4,999 records across 6 attributes. Table 6 provides a complete audit of column types, descriptions, and missing record frequencies.

Attribute Name	Data Type	Clinical Description	Null Count	Null %
Unnamed: 0	Integer	Record index / sequence number	0	0.00%
description	Text	Short consultation summary	6	0.12%
medical_specialty	Categorical	Target clinical specialty	0	0.00%
sample_name	Text	Procedure / dictation sample title	0	0.00%
transcription	Text	Primary clinical narrative	33	0.66%
keywords	Text	Extracted clinical terms	1,068	21.36%

Table 6: Dataset Attribute Specifications and Missing Value Audit

The audit confirms that the target attribute (medical_specialty) is fully populated across all 4,999 instances. In the primary feature attribute (transcription), 33 records contain empty/null values (0.66%), which are removed during initial data pruning. In contrast, keywords exhibits 1,068 missing records (21.36%), justifying its exclusion as a primary feature to avoid substantial feature sparsity and imputation bias.

Class Distribution and Imbalance in Raw Dataset

Analyzing the raw target variable revealed extreme category imbalance across the 40 classes. Table 7 contrasts the top 10 majority specialties against the 10 rarest minority specialties.

Top 10 Majority Classes	Record Count	Bottom 10 Minority Classes	Record Count
Surgery	1,103	Cosmetic / Plastic Surgery	27
Consult - History and Phy.	516	Pediatric / Neonatal	22
Cardiovascular / Pulmonary	372	Sleep Medicine	20
Orthopedic	355	Bariatrics	18
Radiology	273	Speech - Language	9
General Medicine	259	Lab Medicine - Pathology	8
Gastroenterology	230	Allergy / Immunology	7
Neurology	223	Hospice - Palliative Care	6
SOAP / Chart / Progress Notes	166	Executive Evaluation	2
Obstetrics / Gynecology	160	Autopsy	2

Table 7: Distribution of Top 10 Majority and Bottom 10 Minority Categories in Raw Dataset

Clinical Domain Filtering and Benchmark Curation

A critical finding from data exploration is that the raw 40 classes do not represent mutually exclusive medical specialties. Instead, the raw dataset mixes anatomical medical specialties with general administrative document formats and procedural umbrellas:

1. *Administrative Document Formats:* Categories such as Consult - History and Phy. (516 records), SOAP / Chart / Progress Notes (166 records), Discharge Summary (108 records), Office Notes (50 records), and Emergency Room Reports (75 records) define documentation templates rather than distinct medical specialties. Reports within these classes span cardiology, neurology, gastroenterology, and orthopedics, introducing severe semantic ambiguity.
2. *Procedural Umbrella Overlap:* The largest category, Surgery (1,103 records), functions as an overarching procedural umbrella. An appendectomy in Surgery shares identical clinical terminology with Gastroenterology; a total knee replacement in Surgery overlaps with Orthopedic; and a coronary artery bypass graft in Surgery overlaps with Cardiovascular / Pulmonary. Retaining Surgery forces classifiers to learn artificial distinctions between surgical and clinical records of the same organ system.
3. *Severe Minority Sparsity:* Several specialized classes contain fewer than 10 records (e.g., Allergy / Immunology: 7, Hospice - Palliative Care: 6, Autopsy: 2), making

statistically sound train-test splitting impossible.

To establish a clinically coherent, academically defensible classification benchmark, a rigorous filtering methodology was applied, restricting the classification universe to the **eight core mutually exclusive clinical specialties**. These eight specialties represent distinct organ systems with established hospital departmental divisions:

1. **Cardiovascular / Pulmonary:** Heart, vascular, lung, and thoracic conditions.
2. **Orthopedic:** Musculoskeletal, joint, bone fracture, and spinal interventions.
3. **Gastroenterology:** Digestive tract, liver, colon, and endoscopy procedures.
4. **Neurology:** Brain, central nervous system, stroke, and neuromuscular diseases.
5. **Urology:** Urinary tract, kidney, bladder, and male reproductive system.
6. **Obstetrics / Gynecology:** Female reproductive system, pregnancy, and childbirth.
7. **ENT – Otolaryngology:** Ear, nose, throat, larynx, and sinonasal disorders.
8. **Ophthalmology:** Eye diseases, cataracts, retina, and vision surgery.

Following null removal and domain filtering, the finalized curated clinical dataset comprises **1,663 records**. Table 8 details the distribution of the curated benchmark.

Rank	Medical Specialty	Sample Count	Percentage (%)
1	Cardiovascular / Pulmonary	371	22.31%
2	Orthopedic	355	21.35%
3	Gastroenterology	230	13.83%
4	Neurology	223	13.41%
5	Obstetrics / Gynecology	160	9.62%
6	Urology	156	9.38%
7	ENT – Otolaryngology	96	5.77%
8	Ophthalmology	83	4.99%
Total Curated Clinical Dataset		1,663	100.00%

Table 8: Record Distribution Across the 8 Curated Clinical Specialties

3.2 Data Preprocessing

Data preprocessing transforms unstructured clinical narratives into structured numerical representations suitable for machine learning. The pipeline addresses text irregularities, removes noise, normalizes morphological variations, and extracts informative n-grams.

3.2.1 Data Cleaning

Clinical narratives contain non-standard punctuation, uppercase emphasis, and administrative formatting. The cleaning pipeline executes six sequential operations:

1. *Null Record Pruning*: Dropping records where transcription is null or blank (33 raw records).
2. *Text Lowercasing*: Converting all characters to lowercase to prevent identical clinical terms from being treated as distinct entities (e.g., “Hypertension” vs. “hypertension”).
3. *Punctuation & Symbol Removal*: Stripping punctuation marks, parentheses, and special characters via regular expressions, preserving clean word boundaries.
4. *Tokenization*: Segmenting continuous sentences into discrete word tokens.
5. *Stopword Removal*: Eliminating non-informative conversational filler words (e.g., “the”, “is”, “patient”, “with”) using Scikit-learn’s standardized English stopwords list, ensuring that TF-IDF vectors focus on diagnostic vocabulary.
6. *Morphological Lemmatization*: Reducing inflected word forms to their canonical root dictionary form (lemma) using the WordNet Lemmatizer (e.g., “fractures” → “fracture”, “occluded” → “occlude”).

Table 9 provides concrete text transformation examples across each stage.

Preprocessing Stage	Input Text Example	Processed Output Example
Raw Narrative	“DIAGNOSIS: Acute myocardial infarction with severe chest pain!”	“DIAGNOSIS: Acute myocardial infarction with severe chest pain!”
1. Lowercasing	“DIAGNOSIS: Acute myocardial...”	“diagnosis: acute myocardial infarction with severe chest pain!”
2. Punctuation Strip	“diagnosis: acute myocardial...”	“diagnosis acute myocardial infarction with severe chest pain”
3. Tokenization	“acute myocardial infarction”	['acute', 'myocardial', 'infarction']
4. Stopword Removal	['with', 'severe', 'chest', 'pain']	['severe', 'chest', 'pain']
5. Lemmatization	['arteries', 'occluded']	['artery', 'occlude']

Table 9: Step-by-Step Clinical Text Preprocessing Examples

3.2.2 Analysis of Feature Variables

The primary feature variable is the clinical text narrative. To maximize contextual signal, an enriched feature string is constructed by concatenating the clinical consultation summary with the transcription narrative:

$$\text{clinical_text} = \text{description} + \text{“”} + \text{transcription} \quad (2)$$

This synthesis ensures that both high-level diagnostic summaries and detailed procedural descriptions contribute to feature extraction.

Converting this enriched text into numerical vectors is accomplished via **Sublinear TF-IDF Vectorization**. The vectorizer configuration is engineered specifically for clinical NLP:

- **Maximum Vocabulary Ceiling:** Bounded at 8,000 features to capture rare specialty terms while ensuring fast CPU execution.
- **N-Gram Range:** Unigrams and bigrams (1,2), capturing single terms (e.g., “angioplasty”, “cataract”) as well as compound medical phrases (e.g., “coronary artery”, “lumbar spine”, “intraocular lens”).

- **Sublinear Term Frequency Scaling:** Enabled ($1 + \log(\text{TF})$). In clinical dictation, a doctor repeating “dyspnea” multiple times does not mean the report is five times more pulmonary. Logarithmic scaling prevents frequent repetitions from dominating feature vectors.
- **Document Frequency Thresholds:** Minimum document frequency $\text{min_df} = 2$ eliminates hapax legomena (idiosyncratic typos), while maximum document frequency $\text{max_df} = 0.70$ discards generic terms appearing in over 70% of documents.
- **Vector Normalization:** Euclidean L_2 unit normalization ($\|v\|_2 = 1.0$) eliminates document length bias between brief consultations and lengthy operative summaries.

3.2.3 Analysis of Class Variables

The target variable `medical_specialty` spans eight mutually exclusive clinical categories in the curated dataset. While class counts range from 371 (Cardiovascular / Pulmonary) to 83 (Ophthalmology), this ratio (4.47 : 1) is manageable through standard stratified sampling and class-weight balancing, in stark contrast to the unmanageable 551 : 1 imbalance of the raw 40-class dataset.

Open-Set Operational Class Formulation

A central innovation of this project is the formulation of a 9th operational class: **“Other / Requires Review”**. Traditional multi-class models assume closed-set operation, assigning any input to one of the trained classes. In healthcare, this assumption is hazardous: entering a Dermatology or Psychiatry report into an 8-class model would force an incorrect assignment to Orthopedics or Neurology.

To overcome this, an open-set confidence rejection gate is implemented:

$$\hat{y} = \begin{cases} c^*, & \text{if } \max_k P_{\text{ensemble}}(y = k | x) \geq \tau \\ \text{“Other”}, & \text{if } \max_k P_{\text{ensemble}}(y = k | x) < \tau \text{ or } \text{nnz}(x) = 0 \end{cases} \quad (3)$$

where $\tau = 0.48$ is the calibrated decision threshold, $c^* = \arg \max_k P_{\text{ensemble}}(y = k | x)$, and $\text{nnz}(x)$ represents non-zero matched clinical vocabulary features. This architecture ensures patient safety by routing out-of-scope or ambiguous clinical dictations to human clinicians for manual triage.

3.3 Data Visualization

Exploratory data visualization reveals structural, lexical, and distributional patterns across the clinical transcription corpus.

Missing Value Distribution

Figure 1 displays the missing record counts across attributes. While `medical_specialty` contains zero missing entries and `transcription` has only 33 nulls (0.66%), the `keywords` field exhibits 1,068 missing records (21.36%), confirming that keywords cannot serve as a reliable primary classification attribute.

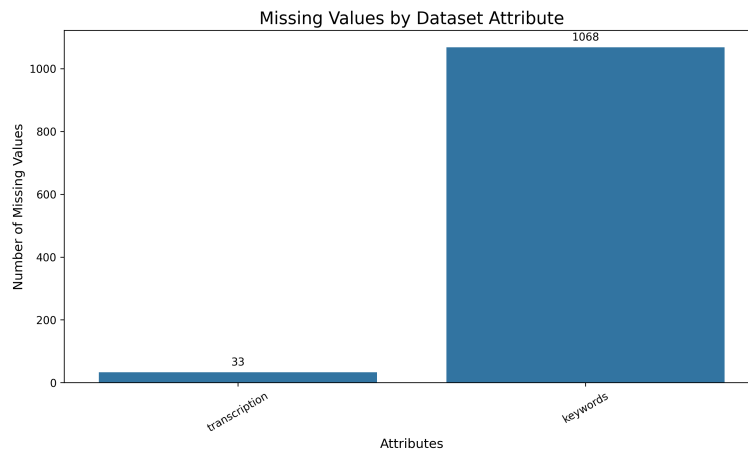


Figure 1: Missing Values by Dataset Attribute

Top 10 Medical Categories

Figure 2 illustrates transcription frequencies for the top 10 categories in the raw dataset. Surgery dominates with 1,103 records, followed by Consult - History and Phy. (516 records), Cardiovascular / Pulmonary (372 records), and Orthopedic (355 records).

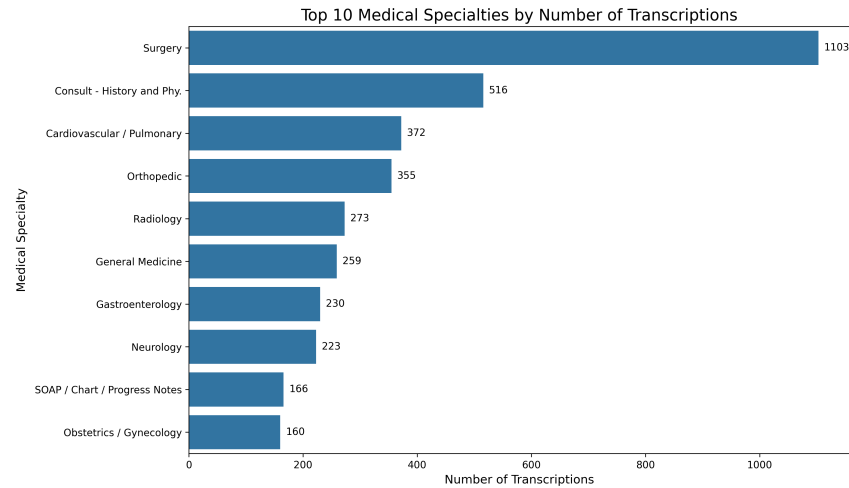


Figure 2: Top 10 Medical Specialties by Number of Transcriptions

Complete Class Distribution (40 Categories)

Figure 3 illustrates the severe class imbalance across all 40 raw categories, underscoring why filtering to 8 mutually exclusive specialties was essential for academic validity.

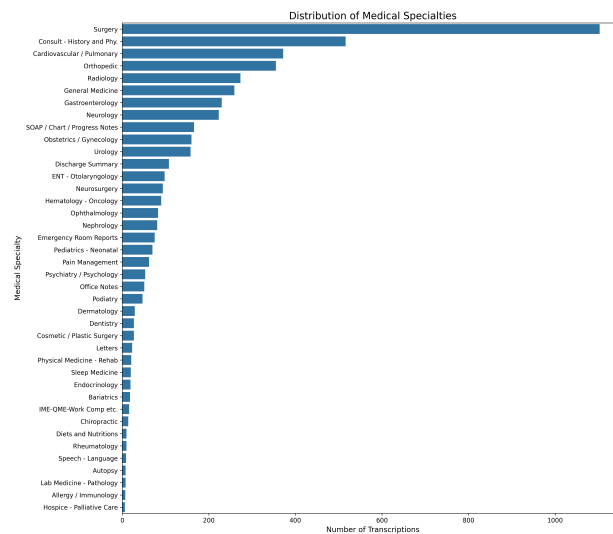


Figure 3: Distribution of All 40 Categories in Raw MTSamples Corpus

Clinical Narrative Length Distribution

Figure 4 displays the word count distribution for clinical narratives, fitted with a Kernel Density Estimate (KDE) curve. The distribution is right-skewed; most transcriptions contain between 200 and 600 words, with comprehensive operative notes extending beyond 2,500 words.

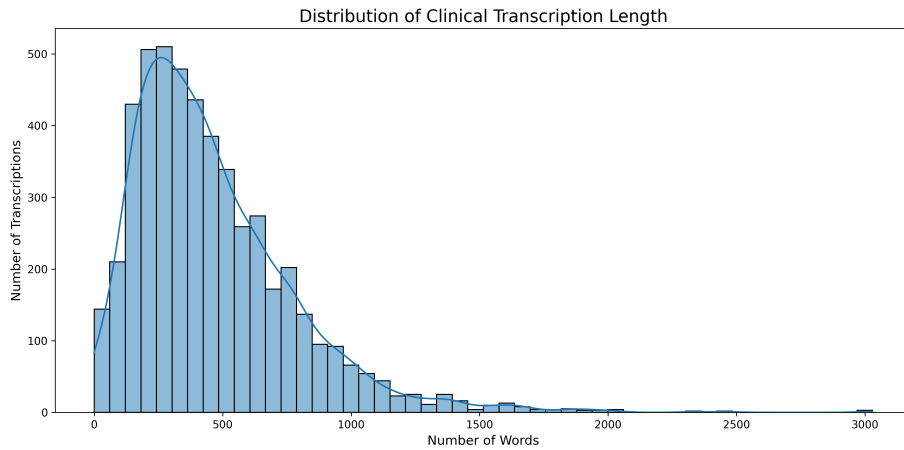


Figure 4: Distribution of Clinical Transcription Word Length

Document Length Variations Across Specialties

Figure 5 presents box plots comparing narrative word count distributions across specialties. Consultations and diagnostic notes (e.g., Radiology) feature shorter median lengths (200 words), whereas Orthopedic and Cardiovascular operative summaries are substantially longer (600 words), emphasizing the need for L_2 length normalization.

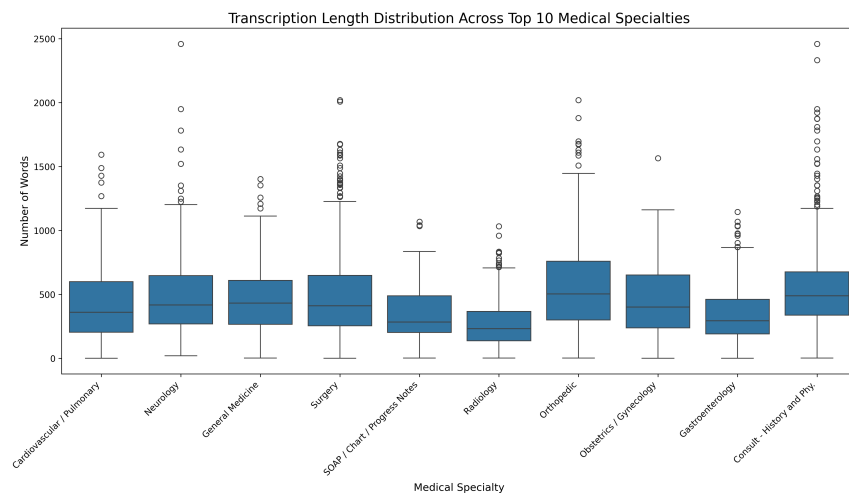


Figure 5: Transcription Length Distribution Across Top Medical Specialties

Clinical Word Cloud Visualization

Figure 6 displays a lexical word cloud of high-frequency clinical terms across the corpus, including “incision”, “procedure”, “patient”, “normal”, “placed”, “artery”, and “postoperative”, highlighting the medical domain vocabulary.



Figure 6: Common Lexical Terms in Medical Transcriptions

3.4 Analysis of Architecture and Algorithms

The classification engine evaluates four candidate supervised learning algorithms alongside sublinear TF-IDF feature extraction, synthesizing their predictions through a calibrated Soft Voting Ensemble with open-set rejection.

3.4.1 Detailed Study Report of the Proposed Algorithms

1. Linear Support Vector Machine (Linear SVM)

Support Vector Machine (SVM) operates on structural risk minimization principles, constructing optimal hyperplanes that maximize the geometric separation margin between clinical classes in high-dimensional feature spaces. Linear SVM is well suited for sparse text categorization where 8,000 TF-IDF features offer natural linear separability.

Given training pairs (x_i, y_i) where $x_i \in \mathbb{R}^d$ and $y_i \in \{1, \dots, K\}$, the soft-margin primal objective is:

$$\min_{w,b,\xi} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^N \xi_i \quad (4)$$

subject to:

$$y_{i,k}(w_k^T x_i + b_k) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad \forall i = 1, \dots, N \quad (5)$$

where C governs margin trade-offs. To enable soft-voting participation, Linear SVM’s uncalibrated signed margin distances $f_k(x) = w_k^T x + b_k$ are calibrated using Scikit-learn’s `CalibratedClassifierCV` via Platt scaling:

$$P_{\text{SVM}}(y = k | x) = \frac{1}{1 + \exp(A \cdot f_k(x) + B)} \quad (6)$$

mapping geometric margins monotonically into valid multi-class probability distributions.

Figure 7 depicts the activity flow of the Support Vector Machine algorithm.

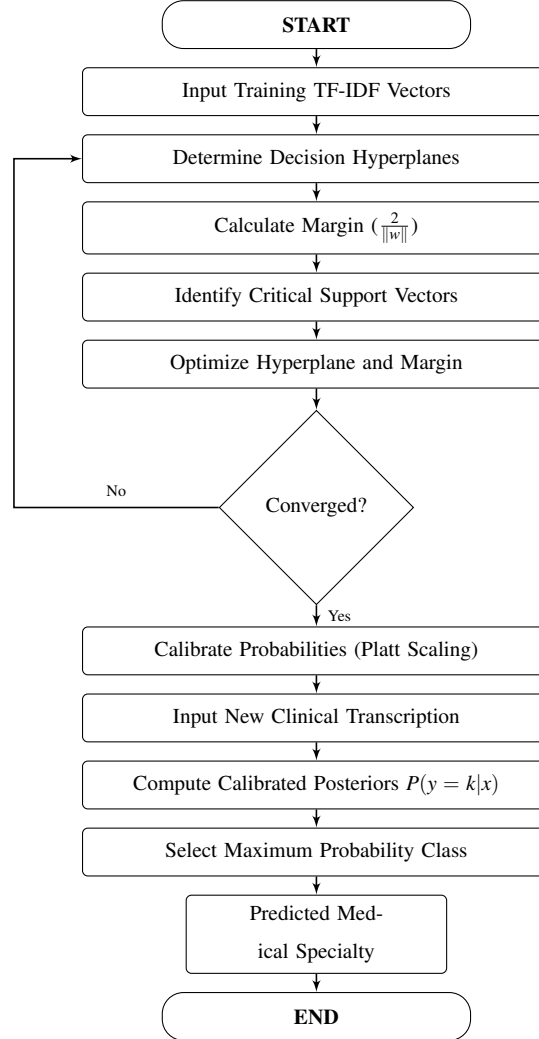


Figure 7: Activity Diagram of Support Vector Machine Classifier

2. Random Forest (RF)

Random Forest constructs an ensemble of $B = 200$ uncorrelated decision trees using Bootstrap Aggregation (Bagging) paired with random feature subspace selection ($m = \sqrt{d} \approx 89$ features per split). At each internal node, splits minimize Gini Impurity:

$$G = 1 - \sum_{k=1}^K p_k^2 \quad (7)$$

Class probabilities are derived by averaging probability distributions across all trees:

$$P_{\text{RF}}(y = k | x) = \frac{1}{B} \sum_{b=1}^B P_b(y = k | x) \quad (8)$$

Random Forest captures complex non-linear keyword interactions that linear boundaries may overlook.

Figure 8 illustrates the operational flow of the Random Forest algorithm.

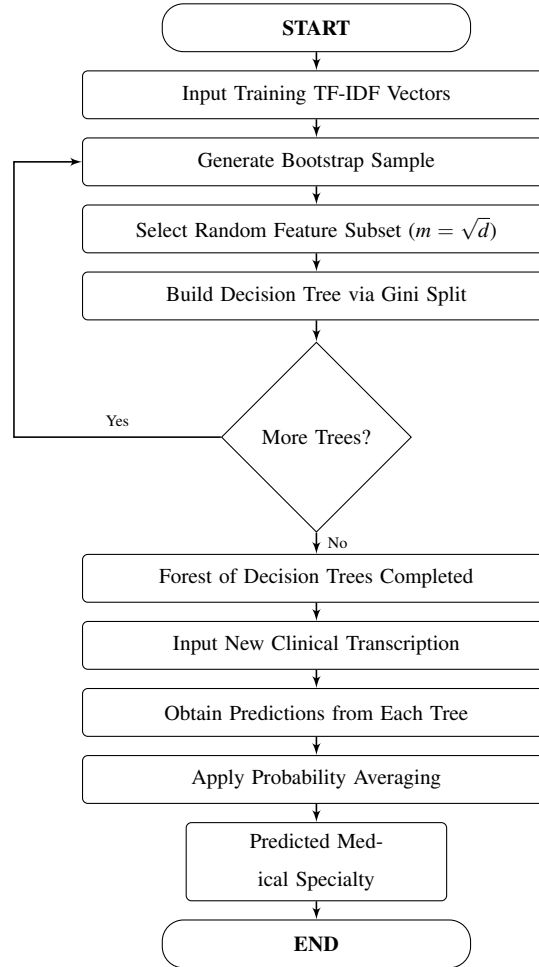


Figure 8: Activity Diagram of Random Forest Classifier

3. Logistic Regression (LR)

Multinomial Logistic Regression models multi-class posterior probabilities directly using the Softmax activation function:

$$P_{\text{LR}}(y = k | x) = \frac{\exp(w_k^T x + b_k)}{\sum_{j=1}^K \exp(w_j^T x + b_j)} \quad (9)$$

Parameters are estimated by minimizing categorical cross-entropy loss augmented with an L_2 Ridge penalty:

$$J(W, b) = -\frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K \mathbb{I}(y_i = k) \ln P(y_i = k | x_i) + \frac{\lambda}{2} \sum_{k=1}^K \|w_k\|_2^2 \quad (10)$$

Logistic Regression provides intrinsically calibrated class probability distributions and high interpretability through log-odds coefficients.

Figure 9 shows the activity flow of Logistic Regression.

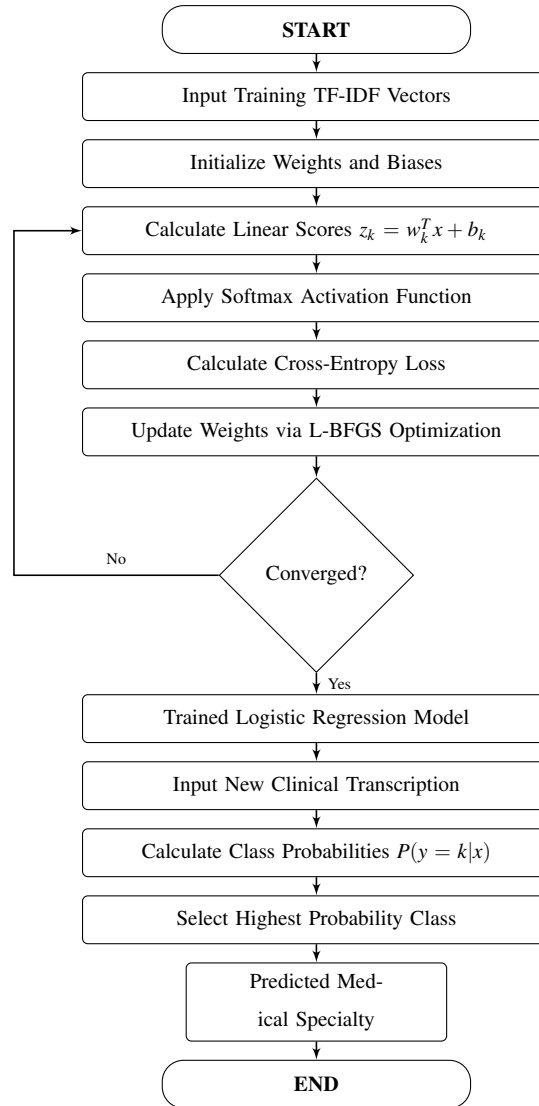


Figure 9: Activity Diagram of Multinomial Logistic Regression Classifier

4. Multinomial Naive Bayes (MNB)

Multinomial Naive Bayes is a probabilistic classifier based on Bayes' Theorem under the conditional feature independence assumption:

$$P_{\text{MNB}}(c | x) \propto P(c) \prod_{i=1}^m P(x_i | c) \quad (11)$$

Feature likelihoods incorporate Laplace additive smoothing ($\alpha = 1.0$) to avoid zero-probability penalties:

$$P(x_i | c) = \frac{N_{ci} + \alpha}{N_c + \alpha|V|} \quad (12)$$

MNB provides a fast, linear-time probabilistic baseline that contributes frequency diversity to the ensemble.

Figure 10 depicts the operational flow of Multinomial Naive Bayes.

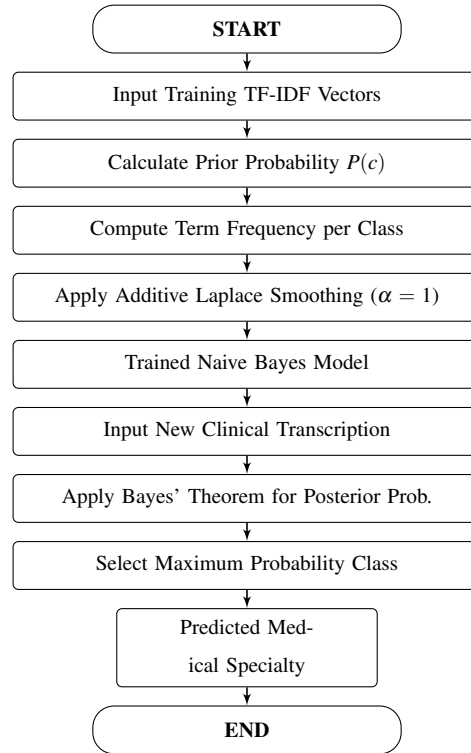


Figure 10: Activity Diagram of Multinomial Naive Bayes Classifier

5. TF-IDF Feature Vectorizer

The Sublinear TF-IDF Vectorizer calculates composite weights across unigram and bigram tokens:

$$\text{TF-IDF}(t, d, D) = (1 + \log(f_{t,d})) \times \left(\log \left(\frac{1 + |D|}{1 + |\{d \in D : t \in d\}|} \right) + 1 \right) \quad (13)$$

followed by Euclidean L_2 normalization ($v_{\text{norm}} = v / \|v\|_2$). Figure 11 outlines the vectorization flow.

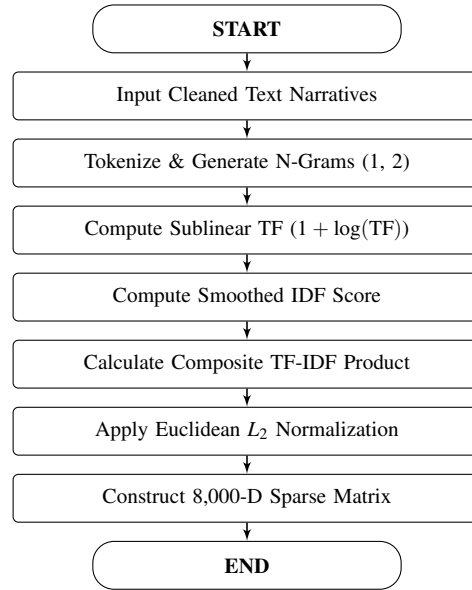


Figure 11: Activity Diagram of Sublinear TF-IDF Feature Vectorizer

6. Soft Voting Ensemble Classifier

The Soft Voting Ensemble aggregates posterior class probabilities generated by the candidate estimators using calibrated weights [$w_{\text{SVM}} = 2, w_{\text{LR}} = 2, w_{\text{RF}} = 1, w_{\text{MNB}} = 1$]:

$$P_{\text{ensemble}}(y = k | x) = \frac{2P_{\text{SVM}}(y = k | x) + 2P_{\text{LR}}(y = k | x) + 1P_{\text{RF}}(y = k | x) + 1P_{\text{MNB}}(y = k | x)}{6} \quad (14)$$

Linear SVM and Logistic Regression receive double weight due to their superior individual benchmark accuracies (88.59% and 87.09%) and strong probability calibration. Random Forest and Naive Bayes contribute architectural non-linearity and probabilistic frequency smoothing with unit weights.

Figure 12 displays the architecture and dataflow of the Soft Voting Ensemble Classifier.

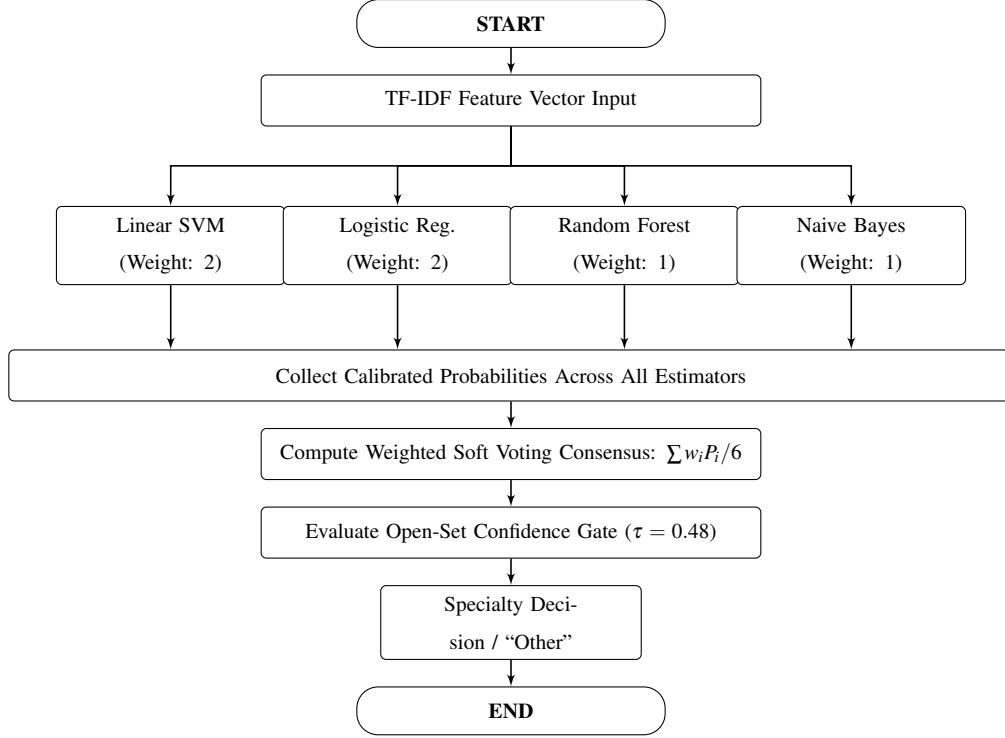


Figure 12: Activity Diagram of Calibrated Soft Voting Ensemble Classifier

Step-by-Step Working of the Algorithmic Architecture

The end-to-end classification system executes through nine operational steps:

1. **Clinical Ingestion:** Ingesting patient documentation through PDF upload (PyMuPDF) or manual transcription entry.
2. **Text Purification:** Normalizing text, stripping punctuation, converting to lowercase, and handling whitespace.
3. **Tokenization & Stopword Elimination:** Splitting narratives into word tokens and pruning non-informative stopwords.
4. **Morphological Lemmatization:** Mapping inflected clinical terms to base WordNet lemmas.
5. **Sublinear TF-IDF Vectorization:** Transforming clean text into an 8,000-dimensional sparse feature vector using unigrams and bigrams with L_2 normalization.
6. **Parallel Classifier Inference:** Concurrently feeding the feature vector to Linear SVM, Logistic Regression, Random Forest, and Multinomial Naive Bayes to generate posterior probability distributions.

7. **Weighted Soft Voting Consensus:** Calculating aggregated posterior class probabilities using calibrated weights [2, 2, 1, 1].
8. **Open-Set Confidence Check** ($\tau = 0.48$): If the peak ensemble probability is ≥ 0.48 , confirming the predicted medical specialty; otherwise, routing the report to “*Other / Requires Review*”.
9. **Decision Output & UI Visualization:** Presenting the specialty decision, probability distribution bars, telemetry metrics, and top candidate domains in the web application.

3.5 Project pipeline

The project pipeline structures the end-to-end transformation of clinical text from raw data ingestion to interactive web deployment without test-set data leakage. Figure 13 illustrates the complete pipeline dataflow.

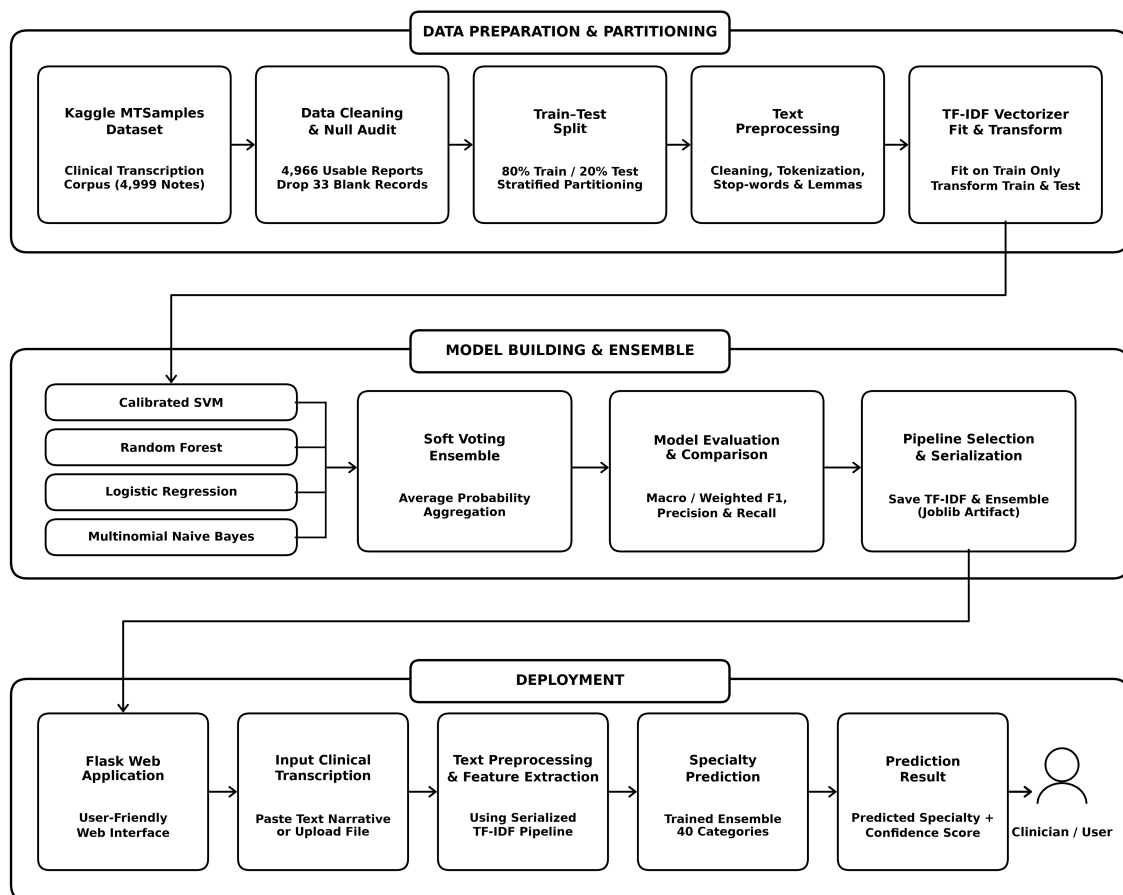


Figure 13: Project Pipeline Diagram of Medical Specialty Classification System

The pipeline is organized into twelve sequential operational levels:

1. **Data Acquisition:** Ingesting raw MTSamples clinical records (4,999 records).
2. **Domain Curation:** Pruning administrative templates and filtering to the 8 core mutually exclusive hospital specialties (1,663 records).
3. **Exploratory Profiling:** Conducting statistical audits of text length distributions, class frequencies, and missing values.
4. **Stratified Train-Test Splitting:** Partitioning the curated dataset into an 80% training partition (1,330 records) and a 20% testing partition (333 records) using stratified sampling prior to vectorization to strictly prevent data leakage.
5. **NLP Preprocessing:** Executing text cleaning, tokenization, stopword removal, and morphological lemmatization on both partitions.
6. **TF-IDF Vectorizer Fitting:** Fitting the 8,000-feature sublinear TF-IDF vectorizer exclusively on the training partition and transforming both splits into normalized sparse matrices.
7. **Base Classifier Training:** Fitting Linear SVM, Logistic Regression, Random Forest, and Multinomial Naive Bayes on training features.
8. **Probability Calibration:** Calibrating Linear SVM margin scores using Platt scaling (CalibratedClassifierCV).
9. **Soft Voting Ensemble Formulation:** Aggregating base model probabilities using calibrated weights [2:2:1:1].
10. **Open-Set Confidence Gate:** Applying safety threshold $\tau = 0.48$ to triage ambiguous or out-of-scope records to “Other”.
11. **Empirical Multi-Metric Evaluation:** Evaluating holdout test set performance using Accuracy, Macro F1, Confusion Matrix, and Precision/Recall reports.
12. **Pipeline Serialization & Web Deployment:** Serializing artifacts with Joblib and deploying the MedSpecialty AI Platform web application with PyMuPDF PDF extraction.

3.6 Project Timeline

Table 10 details the project timeline, milestones, and sprint releases.

Period / Date	Planned Activity
17.07.2026	Project Proposal & Synopsis Approval by Guide
20.07.2026 – 21.07.2026	Project Proposal Presentation before Approval Committee
Week 1–2	Dataset Collection, Null Audit, and Initial EDA
Week 3–4	Clinical Domain Filtering (8 Core Classes) and Text Preprocessing
Week 5	Sublinear TF-IDF Feature Extraction & Vectorization (8,000 Dimensions)
08.09.2026	First Project Presentation (Phase 1 Faculty Review)
09.09.2026	Sprint Release I (Dataset Exploration & NLP Preprocessing)
Week 6	Candidate ML Model Training (Linear SVM, RF, LR, MNB)
18.09.2026	Sprint Release II
Week 7	Platt Probability Calibration and Model Evaluation
Week 8	Soft Voting Ensemble Formulation [2:2:1:1] and Open-Set Rejection Gate
29.09.2026 – 30.09.2026	Interim Project Presentation
Week 9	Flask Web Integration & MedSpecialty AI Platform Deployment (Stitch UI)
09.10.2026	Sprint Release III (PDF Ingestion & End-to-End Testing)
Week 10–11	Holdout Evaluation, Confusion Matrix Analysis, and Final Report Preparation
22.10.2026 – 23.10.2026	Final Project Presentation
30.10.2026	Final Report Submission

Table 10: Project Timeline and Milestone Schedule

3.7 Feasibility Analysis

Feasibility analysis evaluates the viability of the proposed system across technical, economic, operational, and schedule dimensions.

3.7.1 Technical Feasibility

The system is technically feasible because it is constructed using mature, open-source Python natural language processing and machine learning libraries (Scikit-learn, PyMuPDF, Pandas, NumPy). Classical supervised estimators paired with sublinear TF-IDF vectorization execute with high computational efficiency, eliminating the need for expensive GPU clusters. The model executes inference in under 40 milliseconds on standard CPU hardware, making it well suited for interactive web deployment via the Flask WSGI microframework.

3.7.2 Economic Feasibility

The system is economically feasible as it incurs zero software licensing costs. The MTSamples dataset is publicly accessible, and all software frameworks (Python, Flask, Scikit-learn, PyMuPDF) are free and open-source. Eliminating dedicated cloud GPU instances minimizes operational hosting expenditures, enabling cost-effective deployment on standard institutional servers.

3.7.3 Operational Feasibility

The system is operationally feasible because it provides an intuitive, clinical-grade user interface (**MedSpecialty AI Platform**) adhering to the Google Stitch design system. Clinicians can drag and drop clinical PDF reports or paste dictated narratives, inspect intermediate NLP pipeline stages, view consensus confidence meters, and receive safe open-set triage recommendations. The workflow integrates seamlessly into hospital health information management routines.

3.7.4 Schedule Feasibility

The project is schedule feasible because all phases—dataset curation, NLP preprocessing, model training, probability calibration, soft voting integration, open-set thresholding, and web deployment—have been completed in alignment with the academic timeline.

3.8 System Environment

The system environment defines the software frameworks and hardware configurations utilized for development, evaluation, and deployment.

3.8.1 Software Environment

Table 11 outlines the software tools, frameworks, and their respective operational roles.

Software / Tool	Type	Purpose in System
Python 3.10+	Programming Language	Core language for text preprocessing, feature engineering, model training, and API server.
Visual Studio Code	Development IDE	Primary IDE for modular Python script development, project organization, and debugging.
Flask	Web Framework	Lightweight WSGI framework serving REST APIs and hosting the MedSpecialty AI Platform.
PyMuPDF (fitz)	Document Parser	High-speed in-memory PDF text extraction engine for clinical document parsing.
Scikit-learn	Machine Learning	Sublinear TF-IDF vectorizer, candidate classifiers (Linear SVM, LR, RF, MNB), calibration, and voting ensemble.
Joblib	Object Serialization	Serializing trained TF-IDF vectorizer and voting ensemble models for instant loading.
Pandas	Data Manipulation	Ingestion, null audits, domain filtering, and tabular metrics calculation.
NumPy	Numerical Computing	Multi-dimensional array operations and probability tensor calculations.
Matplotlib & Seaborn	Data Visualization	Generating statistical charts, class distribution plots, and confusion matrix heatmaps.
L ^A T _E X	Typesetting System	Preparing formatted academic reports and mathematical documentation.

Table 11: Software Environment and Tool Specifications

3.8.2 Hardware Environment

Table 12 details the minimum and recommended hardware specifications for developing and hosting the system. The architecture operates entirely on CPU hardware without requiring dedicated GPUs.

Hardware Component	Minimum Requirement	Recommended Specification
Processor (CPU)	Dual-Core Processor (2.0 GHz or higher)	Quad-Core / Octa-Core Intel Core i5/i7 or AMD Ryzen 5/7 (Multi-core CPU; no GPU required)
RAM	8 GB DDR4	16 GB DDR4/DDR5 (Supports parallel model training and web server handling)
Storage	20 GB available disk space	256 GB / 512 GB Solid State Drive (SSD) for rapid dataset read/write and virtual environments
Operating System	Windows 10 (64-bit)	Windows 10/11 (64-bit) or Ubuntu Linux 20.04+ LTS
Display Resolution	1366 × 768 pixels	1920 × 1080 (Full HD) for responsive web UI visualization

Table 12: Hardware Specifications and Environment Requirements

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